

第五章

Example

例1: 总体 X 服从期望为 $\frac{1}{\lambda}$ ($\lambda > 0$)的指数分布, $\tilde{X} = (X_1, X_2, \dots, X_n)$ 为来自 X 的简单随机样本.则

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由于独立同分布的中心极限定

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$$Y = \frac{1}{100} \sum_{i=1}^{100} X_i \text{ 近似地 } \sim N\left(\frac{1}{\lambda}, \frac{1}{100\lambda^2}\right),$$

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$$Y = \frac{1}{100} \sum_{i=1}^{100} X_i \stackrel{\text{近似地}}{\sim} N(\frac{1}{\lambda}, \frac{1}{100\lambda^2}),$$

$$\frac{1}{50} \sum_{i=1}^{100} X_i = 2Y \stackrel{\text{近似地}}{\sim} N(\frac{2}{\lambda}, \frac{4}{100\lambda^2}).$$

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事实上,CLT仅是分布类型上的近似,变量的期望和方差是其真实的期望和方差,所以此题其实解答为 $N(\cdot, \cdot)$, 又由于

$$E(\frac{1}{50} \sum_{i=1}^{100} X_i) = \frac{1}{50} \sum_{i=1}^{100} \frac{1}{\lambda} = \frac{2}{\lambda}, \quad D(\frac{1}{50} \sum_{i=1}^{100} X_i) = \frac{1}{50^2} \sum_{i=1}^{100} \frac{1}{\lambda^2} = \frac{1}{25\lambda^2}$$

因此答案为 $N(\frac{2}{\lambda}, \frac{1}{25\lambda^2})$

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$$P\{\frac{1}{100} \sum_{i=1}^{100} X_i^2 \leq \frac{2}{\lambda^2}\} \approx \Phi\left(\frac{\frac{2}{\lambda^2} - \mu}{\sigma}\right),$$

其中

$$\mu = E\left(\frac{1}{100} \sum_{i=1}^{100} X_i^2\right) = E(X_1^2) = \text{Var}(X_1) + (E(X_1))^2 = \frac{2}{\lambda^2},$$

$$\sigma^2 = \text{Var}\left(\frac{1}{100} \sum_{i=1}^{100} X_i^2\right).$$

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答案为 $\frac{1}{2}$.

(3)用 Z 表示 $\{\lambda X_i \leq 1, i = 1, 2, \dots, 100\}$ 发生的次数,则 Z 近似地 $\sim$???

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由CLT知,当 n 很大时,二项分布可以用正态分布近似,因此

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由CLT知,当 n 很大时,二项分布可以用正态分布近似,因此答案是 $N(100(1 - e^{-1}), 100(1 - e^{-1})e^{-1})$.

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数,则 $\frac{\xi}{n} \xrightarrow{P} ???$

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用 ξ 表示 $\{\lambda X_i \leq 1, i = 1, 2, \dots, n\}$ 发生的次

数, 则 $\frac{\xi}{n} \xrightarrow{P} ??? \quad 1 - e^{-1}$.

Example

例2: 总体 X 服从 $U(0, 1)$, $\tilde{X} = (X_1, X_2, \dots, X_n)$ 为来自 X 的简单随机样本.

(1) 记 $Y_n = (X_1 X_2 \cdots X_n)^{1/n}$, 则 $Y_n \xrightarrow{P} ???$

(2) 记 $Z_n = \max_{1 \leq i \leq n} X_i$, 则 $Z_n \xrightarrow{P} ???$ (类似Ch5, Ex4)

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(1) 由于 $\ln(Y_n) = \frac{1}{n} \sum_{i=1}^n \ln X_i \xrightarrow{P} -1$, 其中 $E(\ln X_i) = -1$.

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(1) 由于 $\ln(Y_n) = \frac{1}{n} \sum_{i=1}^n \ln X_i \xrightarrow{P} -1$, 其中 $E(\ln X_i) = -1$.
故 $Y_n \xrightarrow{P} e^{-1}$.

(2)先写出 X_i 的密度函数,从而推出分布函数 $F(x)$;然后再利用 $Z_n = \max_{1 \leq i \leq n} X_i$ 的分布函数为 $F_{(n)}(z) = F^n(z)$,得到 Z_n 的分布函数;再求导得到 Z_n 的密度函数,然后求得 Z_n 的期望($\frac{n}{n+1} \cdot 1$)和方差($\frac{n}{n+2} - (\frac{n}{n+1})^2$), 利用切比雪夫不等式, 可知

$$Z_n - E(Z_n) = Z_n - \frac{n}{n+1} \xrightarrow{P} 0,$$

而 $E(Z_n) = \frac{n}{n+1} \rightarrow 1$, 当 $n \rightarrow \infty$ 时. 故 $Z_n \xrightarrow{P} 1$.

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另: 求 n ——参见上课课件例1(两种方式,切比雪夫不等式和中心极限定理)

第六章

Example

例3: 从总体 $X \sim N(\mu_1, \sigma_1^2)$ 和总体 $Y \sim N(\mu_2, \sigma_2^2)$ 分别抽取了容量为 $n = 6$ 和 $m = 12$ 的两组独立样本 $\tilde{X} = (X_1, X_2, \dots, X_n)$, $\tilde{Y} = (Y_1, Y_2, \dots, Y_m)$. 其样本均值分别记为 $\bar{X}, \bar{Y}$, 其样本方差分别记为 S_1^2, S_2^2 .

$$(1) \sum_{i=1}^6 \frac{(X_i - \mu_1)^2}{\sigma_1^2} \sim ???$$

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$$\chi^2(5);$$

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$$(2) \sum_{i=1}^6 \frac{(X_i - \bar{X})^2}{\sigma_1^2} \sim ???$$

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$$(3) \frac{6(\bar{X} - \mu_1)^2}{S_1^2} \sim ???$$

$$(1) \sum_{i=1}^6 \frac{(X_i - \mu_1)^2}{\sigma_1^2} \sim ???$$

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$$F(1, 5);$$

$$(4) k \frac{\sum_{i=1}^6 \frac{(X_i - \mu_1)^2}{\sigma_1^2}}{\sum_{j=1}^{12} \frac{(Y_j - \bar{Y})^2}{\sigma_2^2}} \sim ???, k = ???.$$

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$$F(6, 11); \frac{11}{6};$$

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$$F(6, 11); \frac{11}{6};$$

(5) 当 $\mu_1 = \mu_2, \sigma_1 = \sigma_2$ 时, 若已

知 $a \frac{\bar{X} - \bar{Y} + b}{\sqrt{5S_1^2 + 11S_2^2}} \sim t(l)$, 则 $a = ???, b = ???, l = ???$

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$$a = 8; b = 0; l = 16;$$

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$$0.5;$$

(6)现再从 X 中抽取一个独立样本 X_7 ,若已知 $\frac{X_7 - \bar{X}}{bS_1} \sim t(r)$,则 $b = ???, r = ???$

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$$b = \sqrt{\frac{7}{6}}; r = 5;$$

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(7) $E(\bar{X}^2 \cdot S_1^2) = ???,$

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$$(7) E(\bar{X}^2 \cdot S_1^2) = ???, = E(\bar{X}^2) \cdot E(S_1^2) = (\mu_1^2 + \frac{\sigma_1^2}{6}) \cdot \sigma_1^2;$$

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$$(8) \ \text{Var}(\sum_{i=1}^6 (X_i - \bar{X})^2 - \bar{X}) = ???,$$

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(8) $\text{Var}(\sum_{i=1}^6 (X_i - \bar{X})^2 - \bar{X}) = ???,$

$$= \text{Var}((n-1)S_1^2 - \bar{X})$$

$$= \text{Var}((n-1)S_1^2) + D(\bar{X}) + 2\text{cov}((n-1)S_1^2, -\bar{X})$$

$$= (\sigma_1^2)^2 \text{Var}\left(\frac{(n-1)S_1^2}{\sigma_1^2}\right) + \text{Var}(\bar{X})$$

$$= \sigma_1^4 \cdot (2(n-1)) + \frac{\sigma_1^2}{n} = 10\sigma_1^4 + \frac{\sigma_1^2}{6};$$

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(9) $P\{S_2^2 = \sigma_2^2\} = ???$

(6) 现再从 X 中抽取一个独立样本 X_7 , 若已

知 $\frac{X_7 - \bar{X}}{bS_1} \sim t(r)$, 则 $b = ???, r = ???$

$$b = \sqrt{\frac{7}{6}}; r = 5;$$

(7) $E(\bar{X}^2 \cdot S_1^2) = ???, = E(\bar{X}^2) \cdot E(S_1^2) = (\mu_1^2 + \frac{\sigma_1^2}{6}) \cdot \sigma_1^2;$

(8) $\text{Var}(\sum_{i=1}^6 (X_i - \bar{X})^2 - \bar{X}) = ???,$

$$= \text{Var}((n-1)S_1^2 - \bar{X})$$

$$= \text{Var}((n-1)S_1^2) + D(\bar{X}) + 2\text{cov}((n-1)S_1^2, -\bar{X})$$

$$= (\sigma_1^2)^2 \text{Var}\left(\frac{(n-1)S_1^2}{\sigma_1^2}\right) + \text{Var}(\bar{X})$$

$$= \sigma_1^4 \cdot (2(n-1)) + \frac{\sigma_1^2}{n} = 10\sigma_1^4 + \frac{\sigma_1^2}{6};$$

(9) $P\{S_2^2 = \sigma_2^2\} = ??? \quad 0.$

Example

例4: 从总体 $X \sim \chi^2(m)$ 中抽取了容量为16的一组独立样本 $\tilde{X} = (X_1, X_2, \dots, X_{16})$, 样本均值记为 $\bar{X}$, 记 B_2 为其二阶中心矩. 则

(1) $E(\bar{X}) = ???$

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(2)

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- (4)

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思考一下, m 的矩法估计量是什么?

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